🎯 Topic: Odds and Odds Ratios in Logistic Regression

General Model Form

The logistic regression model relates the probability of an outcome (p) to the predictors through the log-odds (logit function):

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \cdot \text{var1} + \beta_2 \cdot \text{var2}$$

- **Goal:** Interpret β_1 and β_2 using the **Odds Ratio** (e^{β}).
- **var1:** Quantitative variable.
- **var2:** Qualitative (binary) variable where 0 is the reference level (Level A) and 1 is the comparison level (Level B).

General Interpretation Rules (For Reference)

Type	Coefficient (β)	Odds Ratio (e^{β})	Interpretation
Quantitative (β_1)	0.45	$e^{0.45} \approx 1.57$	The odds multiply by **1.57** for every one-unit increase in var1 (holding var2 constant).
Qualitative (β_2)	1.2	$e^{1.2} \approx 3.32$	The odds for Level B ($\text{var2} = 1$) are **3.32 times** the odds for Level A ($\text{var2} = 0$) (holding var1 constant).

Exercise 1: Project Success

- **Model:** $\text{logit}(p) = -2.5 + 0.45 \cdot \text{HoursStudied} + 1.2 \cdot \text{STEMMajor}$
  - p: Probability of **successful** project completion.
  - HoursStudied: Average hours studied per week (Quantitative).
  - STEMMajor: 1=STEM, 0=Non-STEM (Reference).

1. Interpret the coefficient for **HoursStudied** (0.45) in terms of odds.
2. Interpret the coefficient for **STEMMajor** (1.2) in terms of odds.

Exercise 2: Customer Churn

- **Model:** $\text{logit}(p) = 0.8 - 0.1 \cdot \text{YearsAsCustomer} + 0.5 \cdot \text{PremiumPlan}$
  - p: Probability of a customer **churning** (canceling).
  - YearsAsCustomer: Years as a customer (Quantitative).
  - PremiumPlan: 1=Premium, 0=Standard (Reference).
  1. Interpret the coefficient for **YearsAsCustomer** (-0.1) in terms of odds.
  2. Interpret the coefficient for **PremiumPlan** (0.5) in terms of odds.

Exercise 3: Car Purchase

- **Model:** $\text{logit}(p) = -1.5 + 0.05 \cdot \text{Income} - 0.6 \cdot \text{UsedCar}$
  - p: Probability of **purchasing a new car**.
  - Income: Annual income (in thousands of dollars) (Quantitative).
  - UsedCar: 1=Owns Used Car, 0=Owns New Car (Reference).
  1. Interpret the coefficient for **Income** (0.05) in terms of odds.
  2. Interpret the coefficient for **UsedCar** (-0.6) in terms of odds.

Exercise 4: Website Click

- **Model:** $\text{logit}(p) = 0.2 + 0.02 \cdot \text{TimeOnPage} + 0.35 \cdot \text{MobileDevice}$
  - p: Probability of **clicking** an ad.
  - TimeOnPage: Time spent on page (in seconds) (Quantitative).
  - MobileDevice: 1=Mobile, 0=Desktop (Reference).
  1. Interpret the coefficient for **TimeOnPage** (0.02) in terms of odds.
  2. Interpret the coefficient for **MobileDevice** (0.35) in terms of odds.

Exercise 5: Loan Default

- **Model:** $\text{logit}(p) = -1.0 - 0.3 \cdot \text{CreditScore} + 1.5 \cdot \text{FirstTimeBorrower}$
  - p: Probability of **defaulting** on a loan.
  - CreditScore: Credit score (in hundreds) (Quantitative).
  - FirstTimeBorrower: 1=First-Time, 0=Returning (Reference).
  1. Interpret the coefficient for **CreditScore** (-0.3) in terms of odds.
  2. Interpret the coefficient for **FirstTimeBorrower** (1.5) in terms of odds.